

NFT (Non-Fungible Token)



It's exciting to see how these technologies can solve real-world problems! Let's explore some use cases, and I'll explain the terms along the way.

Use Case 1: Digital University Credentials and On-chain Resumes

Imagine a university wanting to issue degrees and transcripts that are impossible to fake and easy for employers to verify.

Problem: Traditional paper degrees can be forged. Employers need to spend time and resources verifying credentials, and students often have to request transcripts repeatedly.

Solution: The university issues digital degrees and transcripts as **Soulbound Tokens (SBTs)**.

Explanation of Terms:

- **NFT (Non-Fungible Token):** Think of an NFT as a unique digital item, like a one-of-a-kind painting in the digital world. Each NFT has a distinct identity and cannot be replaced by another identical item. In our university example, a traditional NFT could represent a degree, but it could also be bought and sold, which isn't ideal for personal credentials.
- **Blockchain:** Imagine a giant, shared digital ledger like a super-secure, unchangeable Google Sheet, where everyone can see the entries but no one person can change them without everyone else agreeing. This ledger is copied across many computers (**decentralized**). When the university issues a digital degree, it's recorded on this blockchain.
- **Ownership Digitization:** This means taking something you own (like your degree) and turning its proof of ownership into a digital format on the blockchain. Instead of a paper certificate, you have a digital record proving you earned it.
- **Trustless Transfer:** This doesn't mean there's no trust at all. It means you don't need to trust a single middleman (like a bank or a university administration) for the transaction or verification to happen. The blockchain itself, through its rules and cryptography, ensures the integrity. For example, an employer doesn't need to call

cryptography, ensures the integrity. For example, an employer doesn't need to call the university to verify a degree; they can verify it directly on the blockchain.

- **Disintermediation:** This is about cutting out the middlemen. In our university example, the university issues the credential and the employer verifies it directly without needing a third-party verification service.
- **Decentralized Identity (DID):** This is your digital identity that you control, not a company or government. You have a unique identifier on the blockchain that's tied to your digital credentials, but you decide what information to share and with whom.
- **Soulbound Tokens (SBTs):** These are a special kind of NFT that *cannot be transferred* once issued. This is crucial for credentials like degrees. Your university degree SBT is "bound" to your digital wallet (your "soul") and you can't sell or give it away. This ensures that the credential genuinely represents your achievement and reputation.
- **Verifiable Credentials (VCs):** These are digital proofs of attributes or qualifications that can be cryptographically verified. Your digital degree is a Verifiable Credential. It contains information about your degree, and it's issued and signed by the university.
- **Proof of Authenticity:** When the university issues your degree as an SBT on the blockchain, it's proof that the degree is genuine and not a fake. *Issuer*
- **Proof of Ownership:** By holding the SBT in your digital wallet, you have undeniable proof that you are the owner of that digital degree. *Owner*
- **On-chain Resume / Credentials:** Instead of sending a PDF resume, you could share a link to your on-chain resume, which is a collection of your verified SBTs (degrees, certifications, work experience tokens). This makes your resume instantly verifiable and tamper-proof.
- **Identity Verification:** When an employer wants to verify your identity, they can do so by checking the cryptographic signatures on your SBTs, linked to your Decentralized Identity. *cryptographic sign*
D-ID *3rd-Party*
- **Source Verification:** This confirms that the credential actually came from the legitimate issuer (the university in this case) because of the cryptographic signing.
- **Cryptographic Signing:** When the university issues your SBT degree, they digitally "sign" it with a special digital key. This signature is like a super-secure digital stamp that proves who issued the document and that it hasn't been tampered with.
- **Metadata Hashing:** The details of your degree (your name, degree type, graduation date, etc.) are processed through a special mathematical function called a hash. This creates a unique, fixed-length "fingerprint" of that data. If even

Source / Issuer | Owner | CryptSign

Source Verification | Identity Verification

a single letter in the degree details changes, the hash will be completely different, immediately showing that the data has been altered. This hash is then stored on the blockchain, not the full sensitive details.

- **Issuer Verification:** This is the process of confirming that the entity that issued the credential (the university) is indeed who they claim to be. The blockchain and cryptographic signatures help with this.
- **Self-sovereign Identity:** This is the overarching concept that you control your digital identity and data, not some central authority. You choose what information to share and when.

Use Case 2: Supply Chain Transparency and Product Authenticity

Imagine buying a luxury watch or a high-end electronic device and wanting to be absolutely sure it's genuine and not a counterfeit.

Problem: Counterfeit goods are a huge problem, costing industries billions and eroding consumer trust. Tracking a product's journey from manufacturing to the consumer is often opaque.

Solution: Manufacturers mint an NFT for each product, representing its unique identity and tracking its journey on the blockchain.

Explanation of Terms (reiterating some and adding new ones):

- **NFT (Non-Fungible Token):** In this case, each individual watch or device gets its own unique digital token (NFT). This NFT is like its digital birth certificate and passport.
- **Blockchain:** The shared, unchangeable ledger where the watch's journey is recorded.
- **Ownership Digitization:** The digital record on the blockchain now represents the ownership of that specific physical watch. When you buy it, the NFT can be transferred to you, digitally proving your ownership.
- **Trustless Transfer:** When the watch moves from the factory to the distributor, then to the retailer, and finally to you, the transfer of the NFT (and thus, proof of ownership) can happen directly on the blockchain without needing a central intermediary to confirm each step. → why others will verify this txn
- **Disintermediation:** This removes the need for multiple layers of paperwork and third-party verification services in the supply chain, streamlining the process.
- **Proof of Authenticity:** The NFT itself, linked to the product's unique serial number and manufacturing data, serves as irrefutable proof that the product is authentic and not a fake.
- **Proof of Ownership:** When you receive the NFT for your watch, you have digital, verifiable proof that you own that specific watch.

verifiable proof that you own that specific item.

- **Source Verification:** You can trace the NFT back to its origin (the manufacturer) on the blockchain, confirming its legitimacy.
- **Cryptographic Signing:** Each time the product changes hands in the supply chain, the entity handling it can cryptographically sign the transaction on the blockchain, adding to the verifiable history.
- **Metadata Hashing:** Important details about the product (e.g., manufacturing date, materials used, inspection results) can be hashed, and the hash stored on the blockchain, ensuring data integrity without exposing all raw data publicly.
- **Social Proof (Applied to a degree):** While not directly used in the supply chain example, in the context of the university degree, "social proof" would come from the recognition and acceptance of these on-chain credentials by a wide range of employers and other institutions. The more widely adopted, the more "social proof" it has.
- **QR-based Identity Proof:** A QR code on the product's packaging could link to its unique NFT on the blockchain. By scanning the QR code, consumers can instantly see the product's entire history and verify its authenticity.
- **Digital Wrapper:** The NFT acts as a "digital wrapper" around the physical product, containing all its verifiable information and history.
- **Affidavit-backed Minting (Potential use case):** Imagine a situation where a rare antique is being tokenized. A legal affidavit (a sworn statement) could be used as the basis for minting its NFT, ensuring that the initial digital representation is legally sound and backed by a real-world document.
- **Revocable NFTs (Potential use case):** While generally NFTs are immutable, in some specific scenarios, like a temporary license to use software or access a service, a "revocable NFT" could be used. If the license expires or terms are violated, the issuer could revoke the NFT's validity. This is less common for permanent ownership but has niche applications.

These are just a couple of examples, but they highlight the transformative potential of combining these blockchain and identity technologies to create more transparent, secure, and trustless systems in various industries.



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Here's a research plan for that topic. If you need to update it, let me know!

Blockchain Use Cases Explained Simply



Research Websites

- (1) Research and understand the fundamental concepts and typical applications of each term provided in the list.
- (2) Identify common real-world problems that can be addressed or significantly improved by the application of blockchain, digital identity, and tokenization technologies, focusing on issues related to trust, verification, ownership, and data management.
- (3) Select 2-3 distinct and compelling real-life use cases that effectively demonstrate the



Analyze Results



Create Report



Ready in a few mins

Start research



I've completed your research. Feel free to ask me follow-up questions or request changes.



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